

Lab: Communication and Shared Inference

Learning Goal: Analyze how agents align their generative models through communication.

Part 1: Communication Breakdown

Scenario: Describe a time when communication failed (a misunderstanding with a friend, a confusing text message, a cultural miscommunication). Analyze it using Active Inference:

1. What was Person A's generative model / prediction?
2. What was Person B's generative model / prediction?
3. Where did the models diverge?
4. What prediction errors occurred?
5. How was the misunderstanding eventually resolved (if it was)?

Write your response here

Part 2: Theory of Mind Scenarios

Learning Goal: Practice inferring mental states from behavior.

For each scenario, infer the person's likely hidden states (beliefs, desires, feelings) and explain your reasoning:

1. A student arrives late to class, sits in the back, and avoids eye contact with the professor.
2. A friend texts you "It's fine" after you cancel plans.
3. A child covers their eyes and says "You can't see me!"

For scenario 3: What does this reveal about the development of Theory of Mind?

Write your response here

Part 3: Language as Model Alignment

Learning Goal: Analyze how language structures facilitate shared inference.

Exercise: Read the following instruction and draw or describe what it tells you to imagine: "A large red circle is to the left of a small blue square. A green triangle sits on top of the square."

1. How could you verify that another person formed the same mental image from these words?
2. What shared generative model (shared knowledge of shapes, colors, spatial relations) makes this communication possible?
3. Write an instruction that would be ambiguous. Explain why.

Write your response here

Part 4: Echo Chambers and Polarization

Learning Goal: Apply the framework to understand modern communication challenges.

Analysis: Choose a topic where two groups hold strongly opposing views (politics, diet, technology, etc.).

1. Describe each group's generative model (what they expect to be true).
2. Explain why each group tends to seek confirming information (echo chambers).
3. What prediction errors would each group experience if exposed to the other's evidence?

4. Why is it cognitively costly to update one's model in the face of strong counter-evidence?

Write your response here

Part 5: Reflection

In 150 words, reflect: Can two people with very different generative models ever truly understand each other? What would need to happen for genuine understanding to occur?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Failure analysis	Model divergence and prediction error
2	Mental state inference	Theory of Mind
3	Linguistic analysis	Language as generative model alignment
4	Social analysis	Echo chambers and polarization
5	Philosophical reflection	Limits of mutual understanding